Biomimetic Intelligence and Robotics 5 (2025) 100217

Contents lists available at ScienceDirect

Biomimetic Intelligence and Robotics

journal homepage: www.elsevier.com/locate/birob

Research Article

SoftGrasp: Adaptive grasping for dexterous hand based on multimodal imitation learning

Yihong Li, Ce Guo, Junkai Ren *, Bailiang Chen, Chuang Cheng, Hui Zhang *, Huimin Lu

The College of Intelligence Science and Technology, National University of Defense Technology, Changsha 410073, China

ARTICLE INFO

Article history:
Received 25 November 2024
Revised 1 January 2025
Accepted 17 January 2025
Available online 28 January 2025

Keywords:
Adaptive grasping
Dexterous hand
Multimodal fusion
Imitation learning

ABSTRACT

Biomimetic grasping is crucial for robots to interact with the environment and perform complex tasks, making it a key focus in robotics and embodied intelligence. However, achieving human-level finger coordination and force control remains challenging due to the need for multimodal perception, including visual, kinesthetic, and tactile feedback. Although some recent approaches have demonstrated remarkable performance in grasping diverse objects, they often rely on expensive tactile sensors or are restricted to rigid objects. To address these challenges, we introduce SoftGrasp, a novel multimodal imitation learning approach for adaptive, multi-stage grasping of objects with varying sizes, shapes, and hardness. First, we develop an immersive demonstration platform with force feedback to collect rich, human-like grasping datasets. Inspired by human proprioceptive manipulation, this platform gathers multimodal signals, including visual images, robot finger joint angles, and joint torques, during demonstrations. Next, we utilize a multi-head attention mechanism to align and integrate multimodal features, dynamically allocating attention to ensure comprehensive learning. On this basis, we design a behavior cloning method based on an angle-torque loss function, enabling multimodal imitation learning. Finally, we validate SoftGrasp in extensive experiments across various scenarios, demonstrating its ability to adaptively adjust joint forces and finger angles based on real-time inputs. These capabilities result in a 98% success rate in real-world experiments, achieving dexterous and stable grasping. Source code and demonstration videos are available at https://github.com/nubot-nudt/SoftGrasp.

1. Introduction

Dexterous manipulation is a cornerstone skill for robots to interact with the physical world and perform complex tasks [1,2], such as grasping, manipulating, and controlling objects [3,4]. Developing human-level dexterous manipulation, especially basic grasping, is crucial for advancing robotics. While significant progress has been made in mechanical design [5], advanced materials [6], perception [7], and robust control [8] over the past decades, achieving efficient grasping of objects with diverse properties – such as different weights, sizes, shapes, and textures – remains challenging. These difficulties are particularly pronounced when handling deformable objects, where deformations can significantly impact grasping actions and stability [9].

Humans rely on multiple sensory modalities to adaptively adjust their grasping movements when interacting with different objects. These include visually observing deformations, assessing hardness through touch, and recognizing shape and weight via proprioception [10]. However, there is a lack of high-quality multimodal datasets for deformable object grasping, which include data such as images, joint angles, and joint torques. These datasets are essential for serving as blueprints for traditional methods and as models for learning-based approaches. DefGrasp-Sim [11] proposes a large-scale synthetic dataset to evaluate grasping poses for deformable objects. It uses a physical simulator to capture object deformation, although it exhibits systematic variations from real-world characteristics. Existing datasets, such as HMDO [12] and DOT [13], primarily focus on objects themselves, containing RGB images and real-world 3D deformed meshes. However, they lack essential details such as grasping force and contact information.

It is also urgent for robotics to develop approaches that effectively fuse the vastly different sensories and generate feasible grasping actions. Recent advancements in learning-based approaches present promising opportunities for control robots with multimodal sensory information. Such methods excel in

* Corresponding authors.
E-mail addresses: jk.ren@nudt.edu.cn (J. Ren), zhanghui_nudt@126.com (H. Zhang).

https://doi.org/10.1016/j.birob.2025.100217

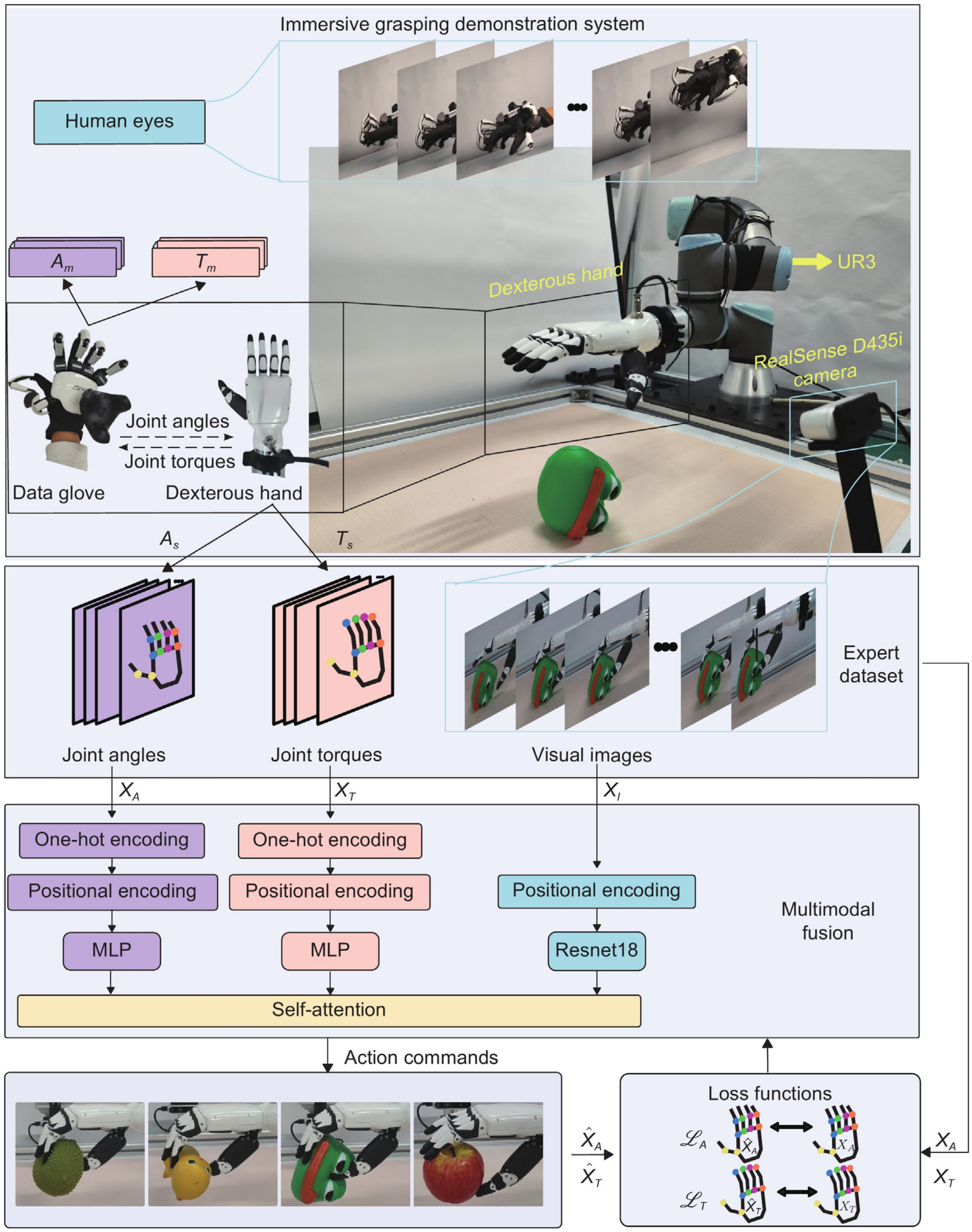

Fig. 1. The structure of SoftGrasp. It uses multisensory human-like grasping data as input, including visual images X_I, finger joint angles X_A, and joint torques X_T. The features of these multimodal data are extracted and fused through a proposed multimodal fusion method. Subsequently, these fused features are converted into action commands for adaptive grasping by the network trained via imitation learning. The multimodal data are collected through an immersive grasping demonstration system, which has a dexterous hand assembled at the end of a UR3 robotic arm, and an Intel RealSense D435i camera fixed on the left side of the arm. It also has a data glove with force feedback function to capture the human grasping actions.

tasks involving complex control of robots with high degrees of freedom [8] and interactive perception [14], which have huge potential for adaptive grasping.

In this work, as shown in Fig. 1, we propose SoftGrasp, a novel multimodal imitation learning framework for adaptive grasping using bionic robotic hands. The objective is to enable robots to grasp objects with different physical properties, such as hardness, shape, and weight, while minimizing object deformation. Our contributions are as follows:

- A novel multimodal imitation learning approach (SoftGrasp) is proposed for adaptive soft grasping. SoftGrasp can effectively learn from real-world multimodal demonstrations to achieve adaptive finger control.

- An immersive human demonstration system with force feedback is designed to enable the collection of diverse humanoid multimodal datasets.
- An efficient multimodal feature fusion mechanism is proposed that dynamically allocates attention to multimodal data, ensuring comprehensive learning.

2. Related work

2.1. Dexterous manipulation

Controlling humanoid dexterous hands remains a challenging problem, with existing methods broadly categorized into traditional control and learning-based methods [15]. Traditional methods often use condition-reaction systems or state-space models to define grasping actions. Condition-reaction methods adapt to grasping under predefined conditions but struggle with objects outside their designed scope, especially those with varying hardness. Similarly, state-space models require precise grasping models [16,17] for numerous finger joints, making them difficult to generalize to objects of diverse shapes, sizes, and materials. These limitations have increased demand for more practical and adaptable control strategies.

The learning-based method has been widely applied in dexterous hand-autonomous operating systems [18,19]. It focus on training control policies from expert demonstrations to replicate dexterous hand control [20,21]. The expert demonstration data is costly to collect. Thus, the researchers present a variety of sources, including physical experiment platforms, simulation environments [22,23], and human demonstration videos [24–26]. These datasets facilitate the learning of complex hand behaviors in high-dimensional action spaces. However, most studies rely on simulated data. The differences in hand-object interaction mechanisms between simulated and real environments may significantly decrease grasping success during the sim-to-real process. Therefore, efficient demonstration data collection in real-world environments is crucial for developing accurate and adaptable control strategies.

2.2. Manipulation of deformable objects

Manipulating deformable objects presents additional challenges due to their unpredictable and dynamic nature [27–29]. Unlike rigid objects, 3D deformable objects can undergo significant changes in shape and size upon contact, complicating grasp planning and control [11,30]. Compared with ropes and fabrics [31], the shape and size of 3D deformable objects will change dynamically with hand contact, further increasing the complexity of constructing accurate models.

While much research has focused on generating grasping poses [32], identifying contact points [33,34], few studies have explored adaptive control strategies that dynamically adjust based on the object's state. For example, Cui et al. [35] propose a method to assess grasping states of deformable objects, but it do not delve into adaptive strategies to handle these objects. Effective manipulation of deformable objects requires a system that not only accommodates varying material properties but also ensures stability and reliability during grasping.

2.3. Multimodal fusion

Robots performing dexterous manipulations encounter various objects, often requiring the integration of multimodal sensory inputs for accurate and adaptive control. Sensory modalities such as vision, touch, and proprioception provide complementary information about the environment. However, integrating these diverse data sources remains a significant challenge.

Traditional multimodal fusion methods concatenate sensory inputs and feed them directly into neural networks [36–38]. While straightforward, this approach struggles to capture complementary relationships and temporal dependencies between modalities, limiting its effectiveness in complex scenarios. Recent studies have explored more advanced methods, for example, Zhuo et al. [39] utilize multimodal inputs, including images and tactile data, for grasping tasks but discretize critical torque information, leading to potential data loss. Ceola et al. [40] employ RGB images, binary tactile information, and proprioceptive information as multimodal inputs. However, the binary tactile information only provides contact states and fails to convey specific details of contact forces, thereby limiting its effectiveness as a reward signal for optimizing the grasping strategy. Building on these advancements, Li et al. [41] apply a multi-attention mechanism to fuse visual, audio, and haptic modalities, significantly improving the efficiency and accuracy of information processing. However, applying such mechanisms to motion-related data, like joint angles and torques, remains underexplored, which is critical for accurate grasping operations.

To address these gaps, this paper proposes a hybrid approach that fuses visual images, joint angles, and joint torques using a multi-head attention mechanism. This approach reduces object deformation during grasping, prevents object slippage, and improves success rates across diverse scenarios.

3. Methodology

In order to achieve efficient adaptive grasping of objects of different sizes, shapes, hardness, and materials, we propose SoftGrasp, an imitation learning method based on multimodal fusion for dexterous hand grasping. We use an immersive grasping demonstration system to capture human manipulation data (Section 3.1), including high-dimensional visual images, low-dimensional finger joint angles, and torques. Then, SoftGrasp extracts the information in these modalities and fuses them flexibly (Section 3.2). Finally, with the imitation learning framework (Section 3.3), the fused features are converted into action commands via neural networks, realizing end-to-end multimodal adaptive grasping actions.

3.1. Immersive grasping demonstration system

We design an immersive grasping demonstration system for collecting human-like grasping data. The system comprises a dexterous humanoid hand, an exoskeleton data glove, and a mapping framework for finger joint angles and torques. By mapping the glove's joint angles to the dexterous hand, the system allows the operator to perform grasping tasks while receiving force feedback from the dexterous hand.

3.1.1. Dexterous hand and data glove

We use a humanoid five-fingered dexterous hand with an anthropomorphic structure that facilitates human manipulation [42]. For grasping tasks, the bending joints of all fingers and the swinging joint of the thumb are mainly used. Moreover, the bending joint motions on the same finger can be coupled and simplified to one finger bending angle represented by one control degree-of-freedom (DoF), as shown in Fig. 2. Thus, the dexterous hand grasping motion can be represented by 6-DoF corresponding to the bending of 5 fingers and the swinging of the thumb, respectively.

As shown in Fig. 2, the data glove provides force feedback by adjusting the stiffness of its exoskeleton joint, enabling the operator to feel fingertip contact forces [43]. In order to make the finger motion of the dexterous hand as close as possible to that of the operator's hand, the data glove should have no less than the simplified control dimensions of the dexterous hand, i.e., 6-DoF.

Table 1
The equation and standard deviation of joint angles mapping function f.

Joint	Mapping function f	Standard deviation
Little Bend	$\hat{A}_{s1} = 0.3678A_{m1}^3 - 1.03A_{m1}^2 + 1.844A_{m1} + 0.037$	9.944×10^{-3}
Ring Bend	$\hat{A}_{s2} = -0.07913A_{m2}^3 - 0.8193A_{m2}^2 + 2.003A_{m2} - 0.02124$	5.903×10^{-3}
Middle Bend	$\hat{A}_{s3} = 0.2115A_{m3}^3 - 1.393A_{m3}^2 + 2.25A_{m3} - 0.0123$	6.201×10^{-3}
Index Bend	$\hat{A}_{s4} = 0.5385A_{m4}^3 - 1.42A_{m4}^2 + 2.064A_{m4} - 0.002128$	5.818×10^{-3}
Thumb Bend	$\hat{A}_{s5} = -196.1A_{m5}^5 + 582A_{m5}^4 - 674.2A_{m5}^3 + 378.3A_{m5}^2 - 99.9A_{m5} + 9.904$	9.721×10^{-3}
Thumb Swing	$\hat{A}_{s6} = -549.1A_{m6}^5 + 1764A_{m6}^4 - 2247A_{m6}^3 + 1417A_{m6}^2 - 438.4A_{m6} + 52.98$	4.446×10^{-3}

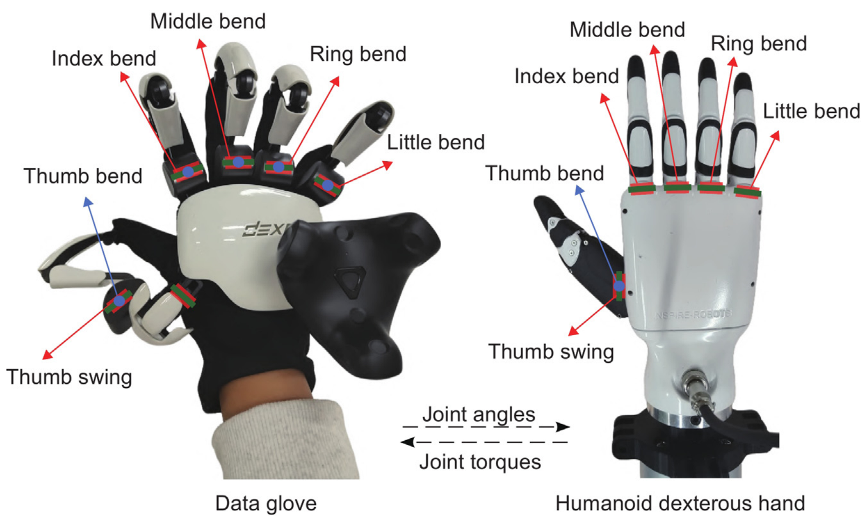

Fig. 2. The immersive grasping demonstration system. The red rectangles at the roots of the index, middle, ring, and little fingers indicate the bending pivot, and the short green line indicates the axis. The red rectangle at the root of the thumb indicates the swinging pivot, and the blue dot indicates the bending pivot.

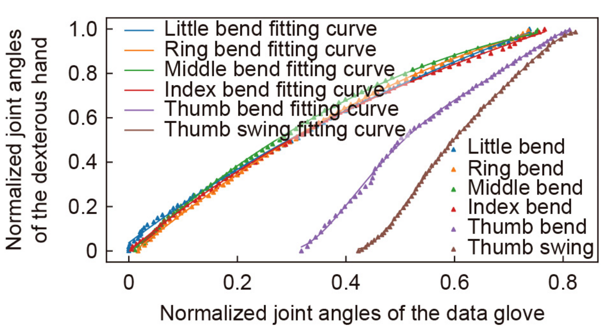

Fig. 3. The relationship of finger joint angles between the data glove and the dexterous hand. The small triangles represent the experimental data. The curves represent the mapping results of the joint angles using the polynomial fitting method.

3.1.2. *Joint angle and torque mapping*

We propose mapping methods for joint angles and joint torques between the data glove and dexterous hand, respectively, realizing effective finger control and fingertip press force feedback.

We use a parameter fitting method to realize the mapping between the actual joint angles of the data glove A_m and the expected joint angles of the dexterous hand A_s. First, we wear the data glove on the dexterous hand and move the fingers of the dexterous hand. The joint angles of the dexterous hand and data glove are recorded and normalized, as shown in Fig. 3. Subsequently, we use the 3-order and 5-order polynomials as the mapping function for each pair of finger joints. The polynomial parameters are computed by fitting the experimental data, as shown in Table 1. Finally, the expected angles of the dexterous hand $\hat{A}_s$ are calculated through the mapping function $\hat{A}_s = f(A_m)$, achieving accurate control of the finger movements.

To simplify the description, we use finger joint torque T_s and T_m to represent the fingertip force of the dexterous hand and the feedback force of the data glove, respectively. The mapping between T_s and T_m is presented as a nonlinear function. The upper limit of the dexterous hand's fingertip press force $T_s^{\max}$ differs from the upper limit of the data glove's feedback force $T_m^{\max}$. A linear mapping from the T_s to T_m would result in untimely and unrealistic force feedback, affecting the immersive feeling of the demonstration. Therefore, we use a segmented function to represent the mapping relationship between T_m and T_s, as in Eq. (1).

$$T_m = \begin{cases} 0, & \text{if } 0 \le T_s < T_s^{\text{con}}, \\ T_s, & \text{if } T_s^{\text{con}} \le T_s < T_m^{\max}, \\ T_m^{\max}, & \text{if } T_m^{\max} \le T_s. \end{cases} \tag{1}$$

During the demonstration, joint angles and torques are fed back to the data glove simultaneously. That makes the object's shape and the dexterous hand's grasping force can be fed back to the operator more accurately. In this regard, we design an immersive grasping control algorithm, as shown in Alg. 1. The algorithm's inputs are the joint torques T_s, joint angles A_s, and joint angles variation ΔA_s of the dexterous hand, as well as the joint angles of the data glove A_m at the current moment. The outputs are the feedback torque $\hat{A}_m$ and feedback angles $\hat{T}_m$ of the data glove, as well as the expected joint angles of the dexterous hand $\hat{A}_s$ at the next moment. We identify the demonstration state of the dexterous hand by the value of joint torques and the variation of joint angles. The demonstration state can be categorized into three types: no contact, contacted but not clutched, and clutched. In the identification basis, T_s^{con} and ΔA_s^{con} refer to the thresholds of joint torques' value and joint angles' variation of the dexterous hand in the contact but not clutched state; T_s^{clu} and ΔA_s^{clu} refer to the thresholds of joint torques' value and joint angles' variation in the clutched state; and μ is the amplifying factor, which is used to regulate the speed of the data glove's joint angles in the contact but not clutched state.

Algorithm 1 Immersive grasping control algorithm

Require: T_s^{con}, T_s^{clu}, ΔA_s^{con}, ΔA_s^{clu}, $T_m^{\max}$, f μ
Input: T_s, A_s, ΔA_s, A_m
Output: $\hat{A}_m$, $\hat{T}_m$, $\hat{A}_s$

while demonstration running **do**
  Refresh: T_s, A_s, ΔA_s, A_m
  if $T_s < T_s^{\text{contact}}$ **then**
    State: no contact
    Dexterous hand: $\hat{A}_s = f(A_m)$
    Data glove: $\hat{T}_m = 0$, $\hat{A}_m = 0$
  else if $T_s^{\text{con}} < T_s < T_s^{\text{clu}}$ **and** $|\Delta A_s| < \Delta A_s^{\text{con}}$ **then**
    State: contact but not clutched
    Dexterous hand: $\hat{A}_s = f(A_m)$
    Data glove: $\hat{T}_m = T_s$, $\hat{A}_m = A_m - \mu|\Delta A_s|$
  else if $T_s > T_s^{\text{clu}}$ **and** $|\Delta A_s| < \Delta A_s^{\text{clu}}$ **then**
    State: clutched
    Dexterous hand: $\hat{A}_s = f(A_m)$
    Data glove: $\hat{T}_m = T_m^{\max}$, $\hat{A}_m = A_m$
  end if
end while

We have now constructed immersive grasping demonstration systems using effective mapping methods for joint angles and torques. We use an exoskeleton data glove to capture the operator's finger joint angles in real-time and control the finger joints of the humanoid dexterous hand. The dexterous hand can execute the actions of the operator's hand accurately and give feedback on the finger joint torque through the data glove. It helps to obtain an expert dataset of human grasping deformable objects with joint angles and torques.

3.2. Multimodal feature representation

With the help of the immersive grasping system shown in Fig. 1, we collect the joint angles $X_A = \{A_1, A_2, \ldots, A_6\}$ and torques $X_T = \{T_1, T_2, \ldots, T_6\}$ of six finger joints on the dexterous hand: little finger bending, ring finger bending, middle finger bending, index finger bending, thumb bending, and thumb swing. Meanwhile, the visual images X_I are captured from the Intel RealSense D435i camera. X_I, X_A, and X_T together constitute the input of the data.

3.2.1. Feature representation

We use the ResNet-18 [44] network to feature vectors (I_1, $\ldots, I_D$) from the visual images X_I.

In this study, since the order of fingers is fixed, we aim to investigate the dependency relationship between each element of the dexterous hand's kinematic state (joint angles and joint torques). To achieve this, we normalize the state variables X_A and X_T, and represent the spatial relationship between finger states using a one-hot coding method. Specifically, we added a 6-dimensional one-hot vector as a positional embedding to each value. Each (1+6)-dimensional state element with a one-hot vector representing its position, is processed through a two-layer multilayer perceptron (MLP) to generate D-dimensional feature vectors with the learned position embeddings $(F_{A1}, F_{A2}, \ldots, F_{AD})$, $(F_{T1}, F_{T2}, \ldots, F_{TD})$ from finger joint angles X_A and torque X_T, respectively.

To further optimize the grasping action of the dexterous hand, we not only consider the state at the t moment but also synthesize the past states for decision-making. Using sinusoidal and cosinusoidal position encodings, we process the states at $t-2$, $t-1$, and t moments. Then, we superimpose these encodings with the past and current state quantities to generate temporal position encodings for the joint angles and torques state.

We integrated the proprioceptive information obtained by dexterous hands during the grasping process, namely joint angles and joint torques, to reflect the basic state of grasping. In order to further enhance the model's ability to process spatial information, we adopted the one-hot encoding technique to encode the state of each finger. In addition, by introducing time series position encoding, we have enhanced the model's ability to process temporal information. We optimized the framework [41] by combining the encoding of temporal and spatial sequences of ontology motion states, improving the system's processing performance for complex sequence data and enhancing the adaptive grasping performance of dexterous hands in complex environments.

3.2.2. Multimodal feature fusion

As shown in Fig. 4, we use a multi-head self-attention mechanism to fuse multimodal features. First, we splice the visual feature vector $(F_{I1}, F_{I2}, \ldots, F_{ID})$, the finger joint angles feature vector $(F_{A1}, F_{A2}, \ldots, F_{AD})$, and the joint torques feature vector $(F_{T1}, F_{T2}, \ldots, F_{TD})$ into a unified matrix $F \in \mathbb{R}^{3\times D}$. Subsequently, we initialize the self-attention mechanism's *Queries*, *Keys*, and *Values* as Eq. (2), where $n = 1, 2, \ldots, H$, H means the number of attention head. $W_n^Q, W_n^K, W_n^V \in \mathbb{R}^{D\times\frac{D}{H}}$ denote the linear transformation matrices for *Queries*, *Keys* and *Values* in the *i*-th group of attention head.

$$Q_n = FW_n^Q,\ K_n = FW_n^K,\ V_n = FW_n^V. \tag{2}$$

We use a Scaled Dot-Product Attention to compute the output of an attention head $head_n$, as shown in Eq. (3), where $d = D/H$, $\sqrt{d}$ denotes the scaling factor. The Softmax layer is used to normalize the output.

$$\begin{aligned} head_n &= \text{Attention}(Q_n, K_n, V_n) \\ &= \text{Softmax}\left(\frac{Q_n K_n^\top}{\sqrt{d}}\right)V_n. \end{aligned} \tag{3}$$

The outputs of all attention heads are concatenated and dot-multiplied by the weight matrix W^O of the attention heads to obtain the fused multimodal feature M, as shown in Eq. (4). $W^O \in \mathbb{R}^{D\times D}$ is a learnable parameter matrix.

$$M = \text{Concat}(head_1, \ldots, head_n)W^O; \tag{4}$$

Finally, we input M into the MLP and predict the dexterous hand joint angles $\hat{X}_A$ and joint torques $\hat{X}_T$ at the next moment.

The multi-head self-attention mechanism can dynamically assign different weights to three modalities, thereby achieving flexible attention allocation. In addition, by incorporating positional encoding into the feature extraction process, this mechanism cannot only effectively integrate information from different time steps but also clearly express the contextual relationships of data in time series. Therefore, in the modal representation of a specific time step, not only are the key features of the current input comprehensively considered, but the relevant information of the previous time step is also fully integrated, thus constructing a temporal feature representation. This multi-head self-attention mechanism that combines dynamic attention allocation and position encoding enhances the model's ability to handle temporal data.

3.3. Imitation learning for SoftGrasp

This paper proposes a behavior cloning approach incorporating an action-torque joint loss function, to facilitate multimodal imitation learning and improve dexterous grasping performance. As shown in Fig. 1, SoftGrasp relies on the expert dataset to

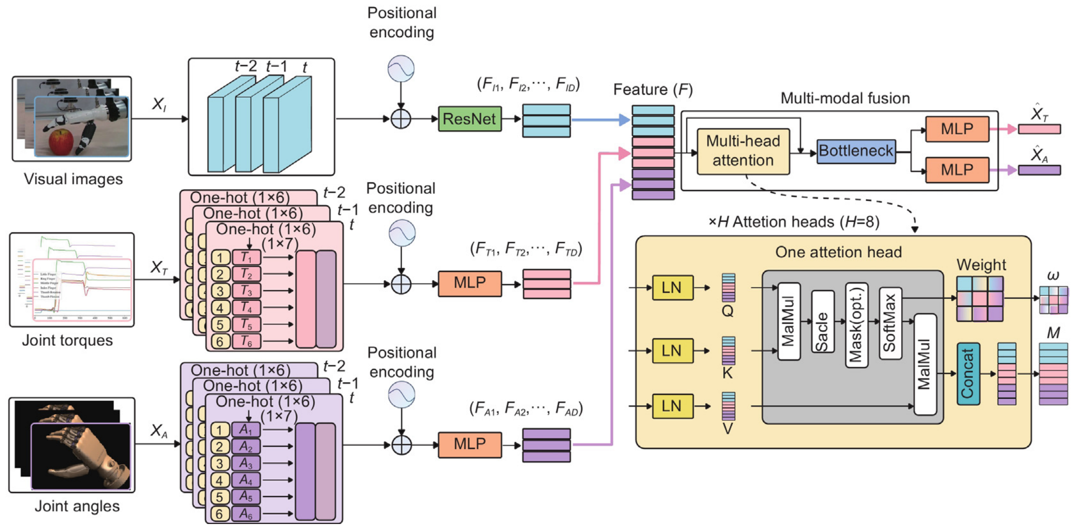

Fig. 4. The network structure of SoftGrasp. Visual images X_I, joint torques X_T, and joint angles X_A are input from the corresponding sensors. These inputs undergo data processing, position encoding, feature extraction, and multimodal fusion steps to obtain fused multimodal features M and attention weights w. The final outputs are the predicted joint torques X_T and joint angles $\hat{X}_A$.

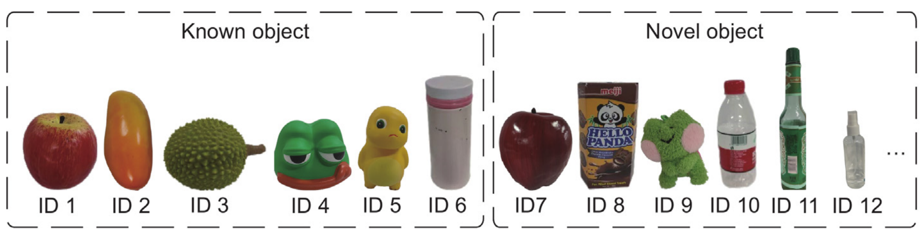

Fig. 5. The objects grasped in experiments. Objects with ID from 1 to 6 are known objects, and objects with ID from 7 to 12 are novel objects.

guide its learning process. Unlike traditional behavior cloning methods, which generally rely on $\mathscr{L}_1$ or $\mathscr{L}_2$ loss functions to fit training data, we developed a composite loss function to enable adaptive control of a dexterous hand. This composite loss function integrates two key components: action loss and torque loss.

The action loss function $\mathscr{L}_A$ and the torque loss function $\mathscr{L}_T$ are defined as the Huber loss between the expected and actual finger joint angles and joint torques, respectively, as shown in the Eqs. (5) and (6). Adjusting the weights of mean square error (MSE) and mean absolute error (MAE) by thresholding δ_A and δ_T.

$$\mathscr{L}_A(X_A, \hat{X}_A) = \begin{cases} \frac{1}{2}(X_A - \hat{X}_A)^2, & \text{if } |X_A - \hat{X}_A| \le \delta_A, \\ \delta_A |X_A - \hat{X}_A| - \frac{1}{2}\delta_A^2, & \text{otherwise;} \end{cases} \tag{5}$$

$$\mathscr{L}_T(X_T, \hat{X}_T) = \begin{cases} \frac{1}{2}(X_T - \hat{X}_T)^2, & \text{if } |X_T - \hat{X}_T| \le \delta_T, \\ \delta_T |X_T - \hat{X}_T| - \frac{1}{2}\delta_T^2, & \text{otherwise.} \end{cases} \tag{6}$$

We combine $\mathscr{L}_A$ and $\mathscr{L}_T$ during training, defined in Eq. (7).

$$\mathscr{L}_{total} = (1 - \lambda)\mathscr{L}_A + \lambda \mathscr{L}_T, \tag{7}$$

where λ is the loss factor used to balance the two loss terms, we can more effectively guide strategy learning through the design of this composite loss function, making it more focused on the influence of torque in grip control.

4. Experiments

In this section, we demonstrate our experiment results in detail and discuss the performance of SoftGrasp. The experiment setups are present in Section 4.1, including grasping objects and training setups. Then, we conduct grasping experiments using SoftGrasp and analyze their characteristic in Section 4.2. Subsequently, we compare and analyze the performance of SoftGrasp with different sensory modalities inputs in Section 4.3. Finally, we compare SoftGrasp with some baseline approaches in success rate and grasping efficiency in Section 4.4, which demonstrate the advancement of our method.

4.1. Experiment setup

All experiments use a 6-DoF UR3 robotic arm equipped with a five-finger dexterous hand, as shown in Fig. 1. The visual images are recorded with a resolution of 480 × 640 pixels at a frequency of 30 Hz by an Intel RealSense D435i camera fixed on the left side of the robotic arm. The joint angles and torques of the dexterous hand are collected at a frequency of 30 Hz.

4.1.1. Grasping objects

The object grasping dataset is constructed through grasping and lifting experiments on 12 objects of different mass, sizes, shapes, and materials. We use a Shore durometer (C-type, range: 0-100HC) to measure the hardness of the objects and classify them into three types: rigid objects (greater than 90HC), soft-plastic objects (10HC-90HC), and deformable objects (less than 10HC). The values for the elastic modulus are obtained as the average of the material properties referenced from a material database MatWeb.[1] The specific parameters are shown in Table 2, and the objects are shown in Fig. 5.

[1] https://matweb.com/index.aspx

Table 2
Parameters of the experimental object. This table lists the various parameters of the objects used in the experiment, including ID, label, mass, Hardness, size, material, elastic modulus, and grasping mode.

ID	Mass (g)	Hardness	Size (mm)	Material	Elastic modulus	Grasp mode
1	15.0	55HC	81 × 81 × 67	MDPE	0.648 GPa	top-down
2	16.4	56HC	70 × 60 × 130	MDPE	0.648 GPa	top-down
3	162.5	<10HC	100 × 80 × 80	TPE	0.204 GPa	top-down
4	76.3	<10HC	110 × 110 × 90	TPE	0.204 GPa	top-down
5	52.0	<10HC	80 × 75 × 120	TPE	0.204 GPa	top-down
6	365.2	>90HC	59 × 59 × 152	Stainless Steel	196 GPa	side
7	16.9	57HC	70 × 70 × 90	MDPE	0.648 GPa	top-down
8	17.3	35HC	81 × 40 × 140	Paperboard	1.935 GPa	top-down
9	43.5	<10HC	115 × 80 × 129	Polyester	5 GPa	top-down
10	17.6	42HC	58 × 58 × 140	PC	2.6 GPa	side
11	389.7	>90HC	45 × 45 × 172	Glass	55 GPa	side
12	37.1	75HC	38 × 38 × 146	PC	2.6 GPa	side

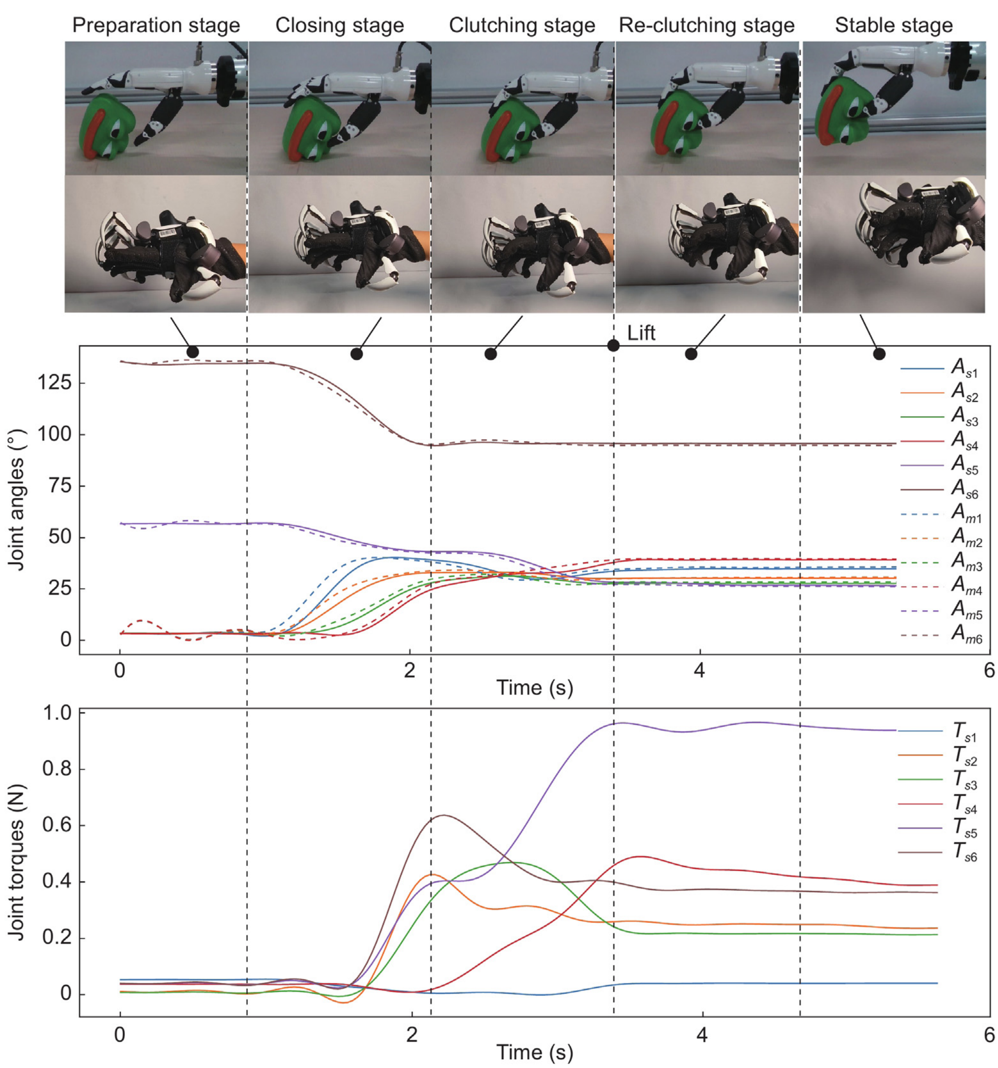

Fig. 6. The process of collecting demonstration datasets. The image above provides visual images of the environment and the object. The middle image, the solid line represents the joint angles of the humanoid dexterous hand($A_{s1}, A_{s2}, A_{s3}, A_{s4}, A_{s5}, A_{s6}$), and the dashed line corresponds to the joint angles of the data glove($A_{m1}, A_{m2}, A_{m3}, A_{m4}, A_{m5}, A_{m6}$). The result demonstrates that the information from the data glove accurately maps to the actions of the dexterous hand. The chart below records the joint torques of the humanoid dexterous hand.

Table 3
The parameter setting for training.

Parameter	Value
Epoch	200
Batch Size	32
Frameskip	5 (frames)
Optimizer	Adam
Loss Factor (λ)	0.1
Number Stack	3 (frames)
Learning Rate (l_r)	$1e^{-4}$

4.1.2. Demonstration data collection

To collect a consistent expert-grasping dataset, the human operator grasped the objects through the immersive grasping system and raised them by 100.0 mm with a speed of 100.0 mm/s. The collected data in one grasping round is shown in Fig. 6, where the top image serves as the input of visual images and provides an intuitive reference for operators. The middle chart indicates the joint torques of the humanoid dexterous hand. The bottom chart depicts the joint angles, showing the consistency of actions between the humanoid dexterous hand and the data glove. According to the characteristics of the grasping process, we can divide the grasping process into the following five stages:

(1) Preparation Stage: The dexterous hand reaches the initial grasping position but has not contacted the object.

(2) Closing Stage: The fingers bend gradually with increasing joint angles. The joint torques rise rapidly once the fingers contact the object.

(3) Clutching Stage: The dexterous hand gradually grasps the object while the finger joint angles and torques tend to stabilize.

(4) Re-Clutching Stage: The object is grasped and lifted off the ground. The finger joint angles and torques adjust slightly.

(5) Stable Stage: The hand holds the object off the ground. The finger joint angles and torques stop adjusting, achieving a new stable stage.

In our dataset, each object has 15 complete grasping trajectories. Each round contains approximately 300 frames of visual images, joint angles, and joint torques. Therefore, one object-grasping dataset consists of approximately 4500 samples. The dataset for training contains 6 objects (ID1-ID6), with a total of 27000 samples.

4.1.3. Training setup

Using Adam as the optimizer, SoftGrasp is trained for 200 epochs with a learning rate $l_r = 1e^{-4}$. Frameskip indicates the number of frames skipped when processing sequence data and Number Stack indicates the number of consecutive frames to be stacked. The specific parameters for training are shown in Table 3. Training and physical demonstration of the robot are conducted using an NVIDIA RTX 3060Ti GPU and an Intel i7-11700F CPU.

4.1.4. Testing setup

We conduct 10 grasping tests on each of the 12 objects shown in Fig. 5. The object's initial position and the robot hands' initial posture are identical in each test for fairness. In each test, the dexterous hand will lift the object vertically by 100 mm after grasping it. If the object falls, the grasp will be considered unsuccessful. Notably, we use fingertip force in Newtons to represent finger joint torque equivalently in these tests. The initial swing angle of the thumb is 125 degrees, showing a decreasing trend during the grasping process.

4.2. Performance analysis

We perform SoftGrasp to grasp soft-plastic, deformable, and rigid objects, as shown in Figs. 7, 8 and 9, respectively. We notice that once the joint torques vary significantly, SoftGrasp immediately slows down the finger bending to prevent over-squeezing, as shown in the Closing Stage, Clutching Stage, and Re-Clutching Stage. However, these stages have significant differences when grasping objects of different hardness. For soft plastic objects (Fig. 7), the Clutching Stage is short, and the joint torques in the Re-Clutching Stage change slightly. For deformable objects (Fig. 8), the joint torques are significantly lower, the Clutching Stage is long, and the torques in the Re-Clutching Stage change obviously. For rigid objects (Fig. 9), the joint torques are high. The Clutching and Re-Clutching stages are performed close to the soft plastic objects. Such behavior highlights the importance of the finger joint torques in dexterous hand grasping. Meanwhile, SoftGrasp can extract the inner information of joint torques, making it suitable for objects with different materials, masses, hardnesses, and shapes.

Furthermore, SoftGrasp can dynamically allocate attention to the modalities at each time step, thereby integrating multisensory data. The multisensory attention layer takes the features of the three modalities, yielding a 3 × 3 raw attention score matrix. Then, at each time step, we sum the attention scores on each modality to obtain the three aggregated attention scores as Eqs. (8), (9), and (10).

$$W_I = \sum_{i=1}^{3} w_{i1}. \tag{8}$$

$$W_T = \sum_{i=1}^{3} w_{i2}. \tag{9}$$

$$W_A = \sum_{i=1}^{3} w_{i3}. \tag{10}$$

where w_{i1}, w_{i2}, w_{i3} represent elements in the raw matrix. The attention scores for each modality, W_I, W_A and W_T, are then normalized together at the same time point, obtaining the normalized attention scores S_I, S_A and S_T.

The normalized attention scores of the three modalities during grasping objects using SoftGrasp are shown at the bottom of Figs. 7, 8 and 9. For the grasping experiment of 3 types of objects, the attention scores of joint torque are highest, the joint angles are medium, and the visual images are lowest. In the initial grasping stage, the model relies more on torque information to ensure the stability and force balance of grasping. As the grasping becomes stable, the scores of joint torque slightly reduce. In contrast, the scores of joint angles increase, and attention gradually shifts more towards joint angles to maintain the grasping posture and perform fine-tuning control. When the grasping reaches the Clutching Stage, the attention scores of the three modalities tend to stabilize. The attention scores for visual images remain relatively stable during the grasping process. This phenomenon reveals that our proposed three sensory modalities are crucial for grasping and play unique roles in every grasping stage.

In summary, the experimental results demonstrate that the characteristics of joint torques and joint angles significantly differ when the dexterous hand grasps deformable and rigid objects. Meanwhile, SoftGrasp can assign priority to the three modalities during grasping and adjust the focus of attention at different stages. Eventually, SoftGrasp realizes adaptive grasping of soft-plastic, deformable, and rigid objects. It also can adaptively adjust the grasping force for objects that deform in grasping.

4.3. Ablation experiment

We set up ablation experiments for Loss Factors and modality, using success rates to evaluate the effect of grasping.

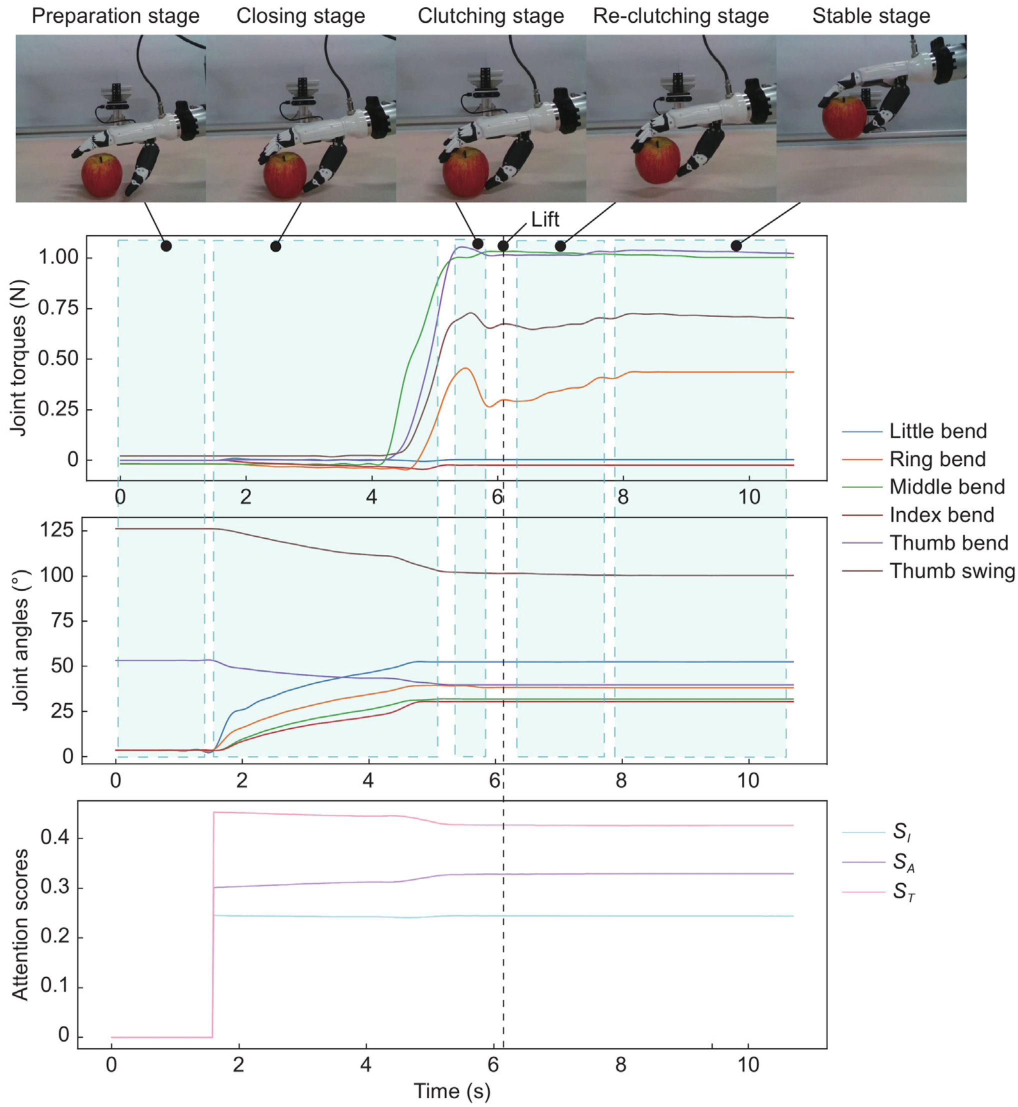

Fig. 7. A single grasping process of a soft-plastic object using SoftGrasp shows the changes in visual images (X_I), joint angles (X_A), joint torques (X_T), and attention scores(S_I, S_A and S_T). The black dotted line in the diagram indicates that the dexterous hand is starting to lift.

Table 4
Success rates of grasping with different Loss Factor l_r.

Loss factor l_r	Avg. succ. rate
SoftGrasp($l_r = 0$)	82/120(68.3%)
SoftGrasp($l_r = 0.05$)	96/120(80.0%)
SoftGrasp($l_r = 0.1$)	**118/120(98.0%)**
SoftGrasp($l_r = 0.15$)	88/120(73.3%)
SoftGrasp($l_r = 0.2$)	86/120(71.1%)
SoftGrasp($l_r = 0.3$)	79/120(65.8%)
SoftGrasp($l_r = 0.5$)	46/120(38.3%)

For Loss Factors, 7 configurations are validated on 12 objects, with 120 trials each in total, and the results are shown in Table 4. It indicates that the highest success rate for grasping is achieved when the Loss Factor is set to 0.1. If the Loss Factor decreases or increases referring to 0.1, the success rate tends to decrease. Specifically, at a Loss Factor of 0.05, the success rate is 80%, and at 0, it is 68.3%. Meanwhile, when the Loss Factor increases to 0.15, 0.2, 0.3, and 0.5, the corresponding success rates are 73.3%, 71.1%, 65.8%, and 38.8%, respectively. Therefore, we use $l_r = 0.1$ to obtain the optimal performance.

Grasping experiments are conducted on 12 objects using SoftGrasp with 7 groups of modalities. For each object, 10 grasping trials are performed, resulting in a total of 120 grasps. The success rates of these experiments are presented in Table 5. SoftGrasp ($X_I + X_T + X_A$) is more adaptable to multiple objects and achieves an average success rate of 98% for grasping all objects. SoftGrasp with different sensory modalities – visual images (X_I), joint angles (X_A), visual images and joint angles ($X_I + X_A$), visual images and joint torques ($X_I + X_T$), and joint torques and angles ($X_T + X_A$) – achieves average success rates of 46%, 34%, 33%, 43%, and 28%, respectively. These results demonstrate that the proposed three modalities are tightly related to grasping. Each modality has utility and performs better when together.

We test the performance of SoftGrasp with different modalities in grasping deformable, soft-plastic, and rigid objects. Visual images of six objects (ID1, ID4, ID5, ID7, ID10, and ID11) during grasping are shown in Fig. 10. From the yellow circle in Fig. 10, SoftGrasps with fewer modalities tend to bend the fingers only to a certain angle. They cannot make delicate and accurate adjustments to the object's size or deformation, especially when grasping small and deformable objects (ID4, ID5, and ID11). Even if the object is successfully grasped, they have difficulty balancing the force of each finger due to the lack of modalities, which leads

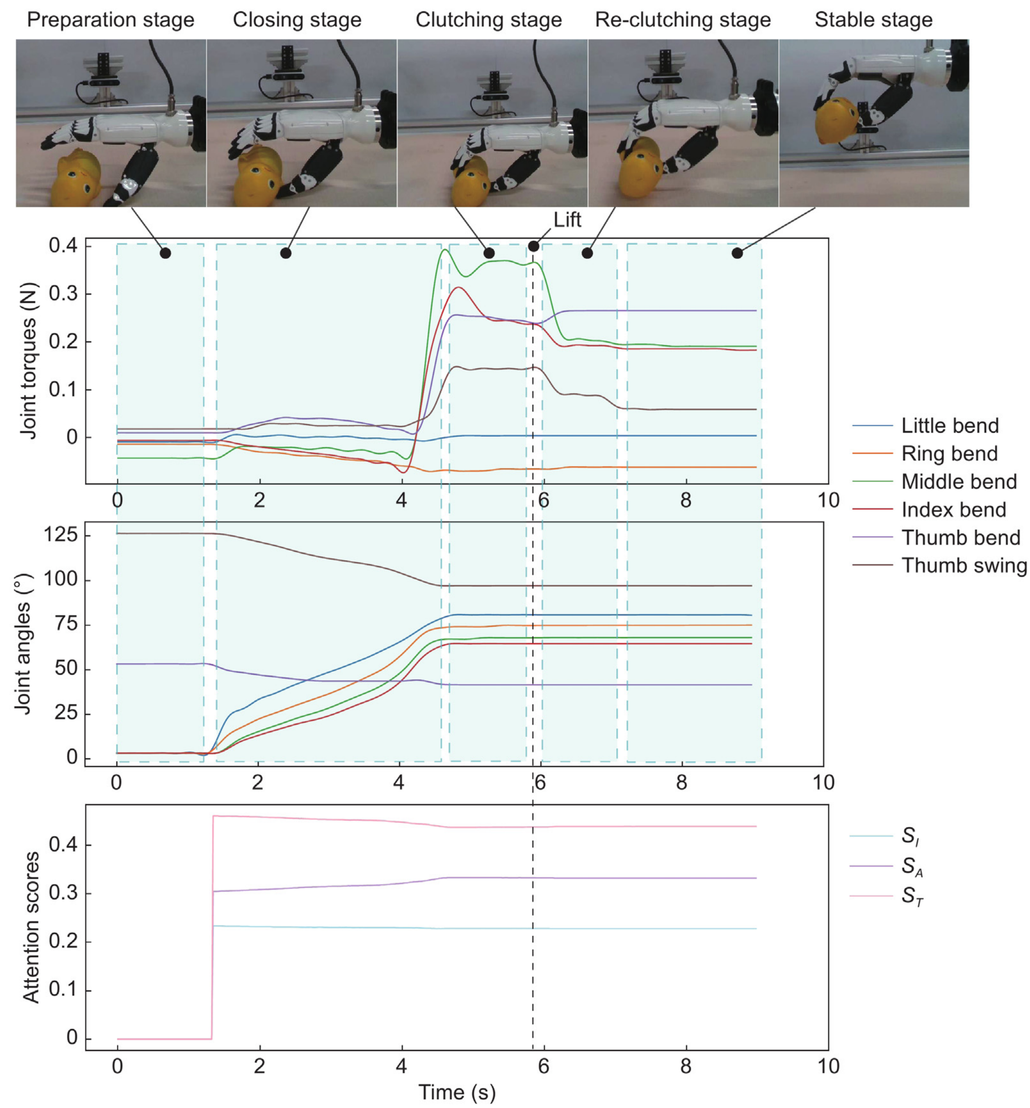

Fig. 8. A single grasping process of a deformable object using SoftGrasp shows the changes in visual images (X_I), joint angles (X_A), joint torques (X_T), and attention scores(S_I, S_A and S_T). The black dotted line in the diagram indicates that the dexterous hand is starting to lift.

Table 5
Success rates of grasping with different modalities.

Modalities	Object ID												Avg. succ.
	1	2	3	4	5	6	7	8	9	10	11	12	rate
SoftGrasp(X_I)	4/10	**10/10**	0/10	**10/10**	9/10	8/10	5/10	0/10	9/10	0/10	0/10	0/10	46%
SoftGrasp(X_A)	**10/10**	0/10	0/10	**10/10**	0/10	1/10	9/10	0/10	9/10	2/10	0/10	0/10	34%
SoftGrasp(X_T)	0/10	0/10	0/10	0/10	0/10	0/10	0/10	0/10	0/10	0/10	0/10	0/10	0%
SoftGrasp($X_I + X_A$)	9/10	**10/10**	0/10	9/10	0/10	1/10	8/10	2/10	0/10	0/10	1/10	0/10	33%
SoftGrasp($X_I + X_T$)	**10/10**	0/10	0/10	**10/10**	0/10	9/10	**10/10**	0/10	**10/10**	3/10	0/10	0/10	43%
SoftGrasp($X_T + X_A$)	3/10	0/10	0/10	**10/10**	0/10	**10/10**	3/10	0/10	8/10	0/10	0/10	0/10	28%
SoftGrasp($X_I + X_T + X_A$)	**10/10**	**10/10**	**10/10**	**10/10**	**10/10**	**10/10**	**10/10**	8/10	**10/10**	**10/10**	**10/10**	**10/10**	**98%**

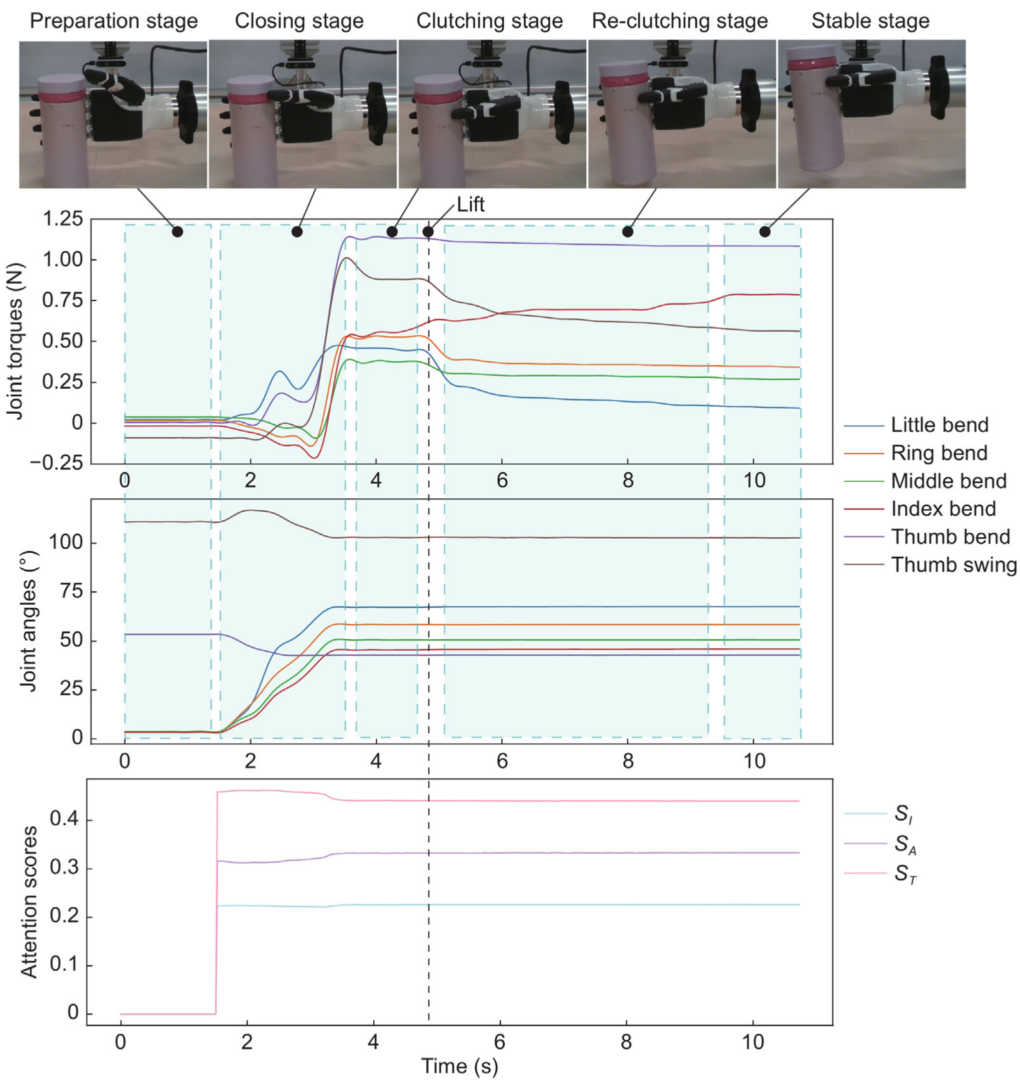

Fig. 9. A single grasping process of a rigid object using SoftGrasp shows the changes in visual images (X_I), joint angles (X_A), joint torques (X_T), and attention scores (S_I, S_A and S_T). The black dotted line in the diagram indicates that the dexterous hand is starting to lift.

Table 6
Success rates of grasping with different strategies.

Strategies	Object ID												Avg. succ.
	1	2	3	4	5	6	7	8	9	10	11	12	rate
SoftGrasp	**10/10**	**10/10**	**10/10**	**10/10**	**10/10**	**10/10**	**10/10**	8/10	**10/10**	**10/10**	**10/10**	**10/10**	**98%**
Direct Connection [37,38]	4/10	0/10	0/10	3/10	0/10	0/10	6/10	0/10	3/10	0/10	0/10	0/10	13%
ACT [45]	**10/10**	9/10	**10/10**	**10/10**	**10/10**	**10/10**	**10/10**	**10/10**	9/10	**10/10**	7/10	4/10	91%
Timesformer [46]	**10/10**	6/10	5/10	9/10	9/10	8/10	9/10	**10/10**	**10/10**	9/10	7/10	3/10	79%

to a low success rate in grasping ID4, ID5, and ID11.

Taking ID1, ID4, and ID6 as examples, the distribution of joint torque in the Stable Stage when grasping successfully for SoftGrasp with different modalities are shown in Fig. 11. SoftGrasp, with three modalities, has small and concentrated joint torques compared to SoftGrasp, which has fewer modalities. It prefers not to use the little finger in grasping, so the little finger has the lowest torque in grasping all three objects. In contrast, the other methods are hard to determine which finger to use in grasping, so they have a larger distribution range of the joint torque for all fingers.

Therefore, SoftGrasp with three modalities performs better than fewer modalities. It can balance joint torques during grasping and adapt to objects of different sizes and hardness.

4.4. Comparison experiment

We compare SoftGrasp with the following baselines:

(1) Direct Connection: A baseline method directly connects the feature vectors extracted from each modality. It is a representative baseline method in [37,38].

(2) ACT [45]: We keep the original backbone of ACT's structure, replacing inputs with our visual images and joint angles, which are aligned with SoftGrasp. The joint torques are not considered in the original ACT algorithm.

(3) Timesformer [46]: We keep the original backbone of Timesformer's structure, replacing inputs with visual images, joint angles, and joint torques representations aligned with SoftGrasp.

Fig. 10. Comparison of dexterous grasping poses in the stable stage of Softgrasp with different modalities. Each row represents the grasping results of different modalities for the same object. The first column (blue box) shows the stable grasping pose achieved by SoftGrasp. The red box in the figure shows the failed grasping pose. For objects ID10 and ID11, we provide the top view image to show the details of finger closure. The yellow circle in the picture indicates that the finger does not contact the object, which may lead to failed grasping.

Grasping experiments are conducted using the three baseline methods and SoftGrasp. For each object, ten grasping trials are performed, amounting to a total of 120 grasps. The success rates of these experiments are presented in Table 6. The visual images of grasping 12 objects are shown in Fig. 12. Taking ID1, ID4, and ID6 as examples, we draw a box diagram of joint torques in the Stable Stage with four different grasping approaches, as shown in Fig. 13.

The results demonstrate that the average success rate of SoftGrasp is 85% higher than the Direct Connection method, 19% higher than Timesformer, and 7% higher than ACT. SoftGrasp demonstrates superior adaptability to small bottle-like objects ID11 and ID12, which are not part of the training data, a unique advantage over Timesformer and ACT. ID11 is a heavy glass bottle that requires a larger force to grasp from the side. ID12 is a small, light plastic bottle that easily slips out of the hand. These bottles challenge the algorithm's ability to adapt to unknown objects.

For the Direct Connection, it executes a similar finger posture to grasp most objects, as seen in Fig. 12. The distribution of joint torques is unstable between objects of different sizes. That means the Direct Connection cannot adapt to objects of different sizes and shows that the feature representation for modalities in SoftGrasp has a significant positive effect.

For ACT, it performs a closure success rate to SoftGrasp, but has much higher joint torques, as shown in Fig. 13. The original ACT algorithm does not treat joint torques as input. It grasps the object at a very fast joint speed as tight as possible. That may change the contact finger of the object, increasing the distribution

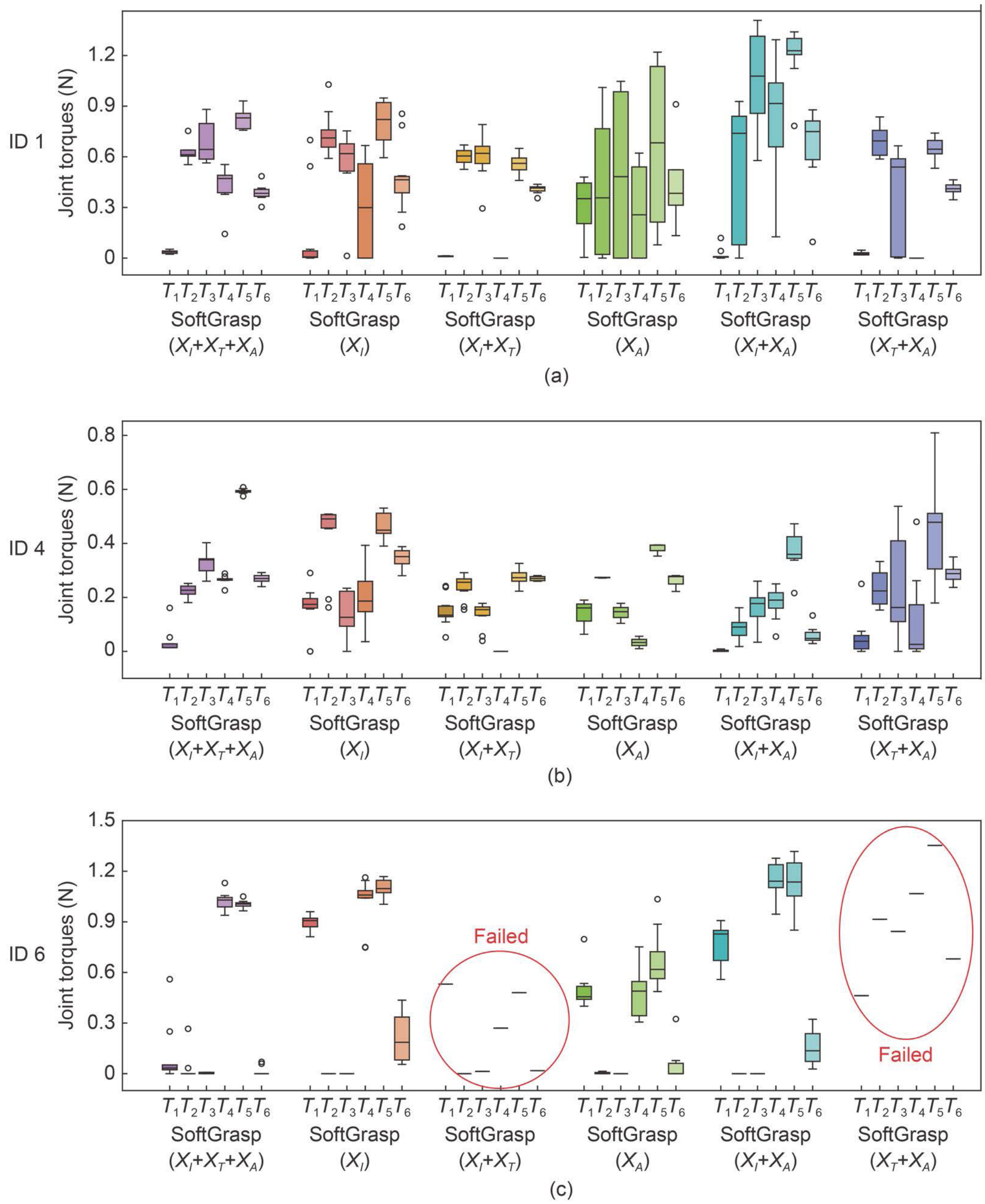

Fig. 11. Distribution of joint torques at the Stable Stage with different modalities of SoftGrasp. For the ID6, two modalities, SoftGrasp (X_I+X_T) and SoftGrasp (X_T+X_A), are marked as failed. That is because the success rate of grasping the ID6 with these two modalities is only 0.1.

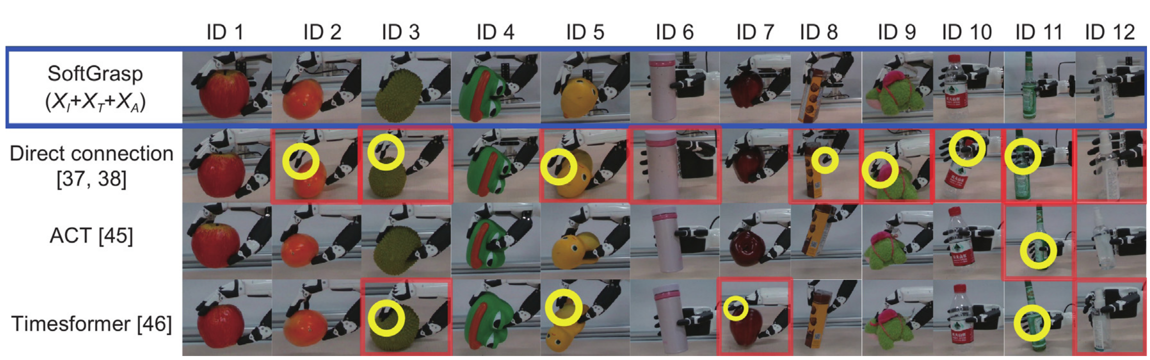

Fig. 12. Comparison plots of dexterous grasping poses in the Stable Stage for different strategies. Each row represents the grasping posture of the same strategy for 12 objects. The first line shows the stable grasping posture implemented by SoftGrasp, and the red box in the figure shows the dexterous hand grasping posture when grasping fails. The yellow circle in the picture indicates that the finger is not in contact with the object, which may lead to failed grasping.

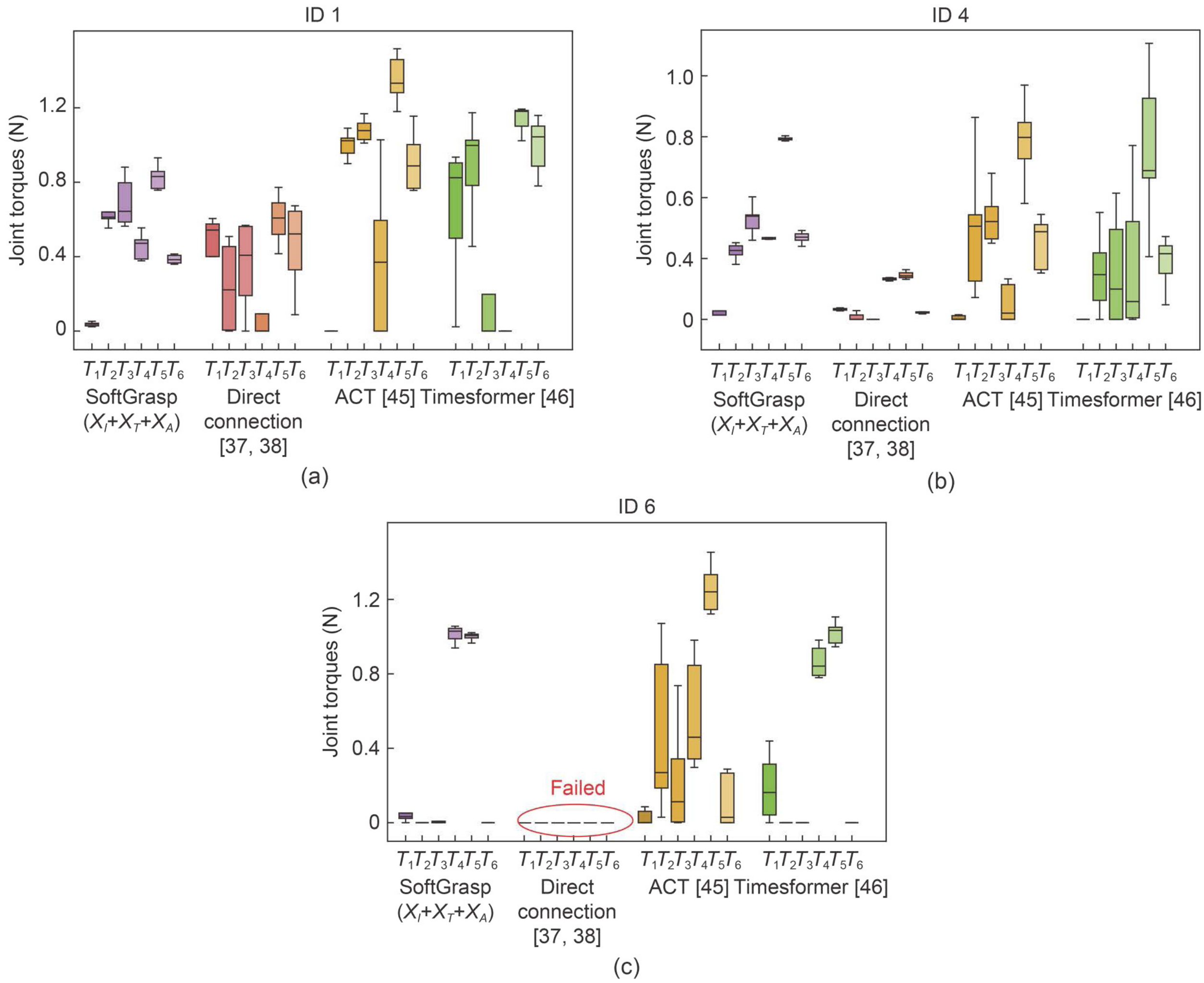

Fig. 13. Distribution of joint torques at the Stable Stage with different strategies. The Direct Connection has a 0% success rate in grasping ID6, so its distribution of joint torques is not plotted and marked as a failed.

Table 7
The average value and standard deviation of inference time for each step in a grasping process with different strategies.

Strategies	Time
SoftGrasp	3.9 (0.7) ms
Direct Connection [37,38]	3.2 (0.6) ms
ACT [45]	12.8 (1.0) ms
Timesformer [46]	16.4 (1.5) ms

of joint torques.

Timesformer has both similar success rates and upper limits of grasping joint torques with SoftGrasp, as shown in Fig. 13. However, its distribution of joint torques is dispersive, representing the lack of ability to control the grasping torque. That shows the importance of feature extraction and fusion because SoftGrasp has treated the features for each modality specifically, while Timesformer does not.

The inference time and standard deviation for each step in a grasping process with different strategies are shown in Table 7 Among them, the Direct Connection strategy exhibits the shortest inference time, but its grasping performance is significantly bad, with a grasping success rate of only 13% (as shown in Table 6). Therefore, it is unsuitable for high-precision and high-reliability grasping tasks. SoftGrasp takes 3.9 ms on average to inferencing and exhibits excellent grasping performance with a success rate of up to 98%. That significantly demonstrates the superiority of our proposed SoftGrasp strategy in grasping efficiency. Although the Timesformer and the ACT strategies have shown good performance in grasping success rates, their inference time is 3∼4 times longer than SoftGrasp, increasing the requirement of compute resources and threshold of deployment on robots.

5. Conclusion

This study proposes **SoftGrasp**, a novel multimodal imitation learning approach designed to learn adaptive finger control from real-world multimodal demonstrations. This method effectively enables multi-stage grasping of objects with varying sizes, shapes, and hardness. Humanoid multimodal demonstration samples are collected through the immersive grasping demonstration system with force feedback, including visual images, finger joint angles, and joint torques. Then, a multi-head attention mechanism is introduced to effectively align and fuse multimodal sensory data, ensuring comprehensive learning across time and modalities. After that, a behavior cloning framework is developed that utilizes fused features to generate adaptive control commands, enabling stable and dexterous grasping. By combining positional and one-hot coding, the model can better understand the robot state feature and learn the intrinsic relationships of individual modalities in different frames.

Extensive experiments demonstrate that SoftGrasp achieves a 98% success rate in real-world tests, significantly outperforming baseline methods. Its dynamic allocation of attention to sensory modalities enables it to adapt to diverse grasping scenarios, including previously unseen objects. Additionally, SoftGrasp's computational efficiency ensures its suitability for real-time robotic applications.

We employ imitation learning to obtain expert policies, which necessitates human demonstrations. However, the demonstrations may not be optimal, and the model performance is sensitive to the quality and quantity of demonstration trajectories and samples. Therefore, our future studies will consider the generative models and reinforcement learning method to reduce the dependency on large datasets and enhance the overall performance of the learned policies.

CRediT authorship contribution statement

Yihong Li: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ce Guo:** Writing – review & editing, Investigation. **Junkai Ren:** Writing – review & editing, Supervision, Conceptualization. **Bailiang Chen:** Writing – original draft, Formal analysis, Data curation. **Chuang Cheng:** Resources, Validation. **Hui Zhang:** Supervision, Resources. **Huimin Lu:** Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work is supported by the Innovation Science Foundation of National University of Defense Technology, China (24-ZZCX-GZZ-11), and the National Science Foundation of China (62373201).

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.birob.2025.100217.

References

[1] M.A. Roa, M.J. Argus, D. Leidner, C. Borst, G. Hirzinger, Power grasp planning for anthropomorphic robot hands, in: Proc. of the IEEE Intl. Conf. on Robotics & Automation, ICRA, 2012, pp. 563–569, http://dx.doi.org/10.1109/ICRA.2012.6225068.

[2] Y. Huang, D. Fan, H. Duan, D. Yan, W. Qi, J. Sun, Q. Liu, P. Wang, Human-like dexterous manipulation for anthropomorphic five-fingered hands: A review, Biomim. Intell. Robot. (2025) 100212, http://dx.doi.org/10.1016/j.birob.2025.100212.

[3] T. Chen, J. Xu, P. Agrawal, A system for general in-hand object re-orientation, in: Proc. of the Conf. on Robot Learning (CoRL), 2022, pp. 297–307, https://proceedings.mlr.press/v164/chen22a.html.

[4] Y. Chen, Y. Geng, F. Zhong, J. Ji, J. Jiang, Z. Lu, H. Dong, Y. Yang, Bi-DexHands: Towards human-level bimanual dexterous manipulation, IEEE Trans. Pattern Anal. Mach. Intell. (TPAMI) 46 (5) (2024) 2804–2818, http://dx.doi.org/10.1109/TPAMI.2023.3339515.

[5] Q. Hu, E. Dong, D. Sun, Soft gripper design based on the integration of flat dry adhesive, soft actuator, and microspine, IEEE Trans. Robot. (TRO) 37 (4) (2021) 1065–1080, http://dx.doi.org/10.1109/TRO.2020.3043981.

[6] S. Xu, Y. Zhang, L. Jia, K.E. Mathewson, K.-I. Jang, J. Kim, H. Fu, X. Huang, P. Chava, R. Wang, S. Bhole, L. Wang, Y.J. Na, Y. Guan, M. Flavin, Z. Han, Y. Huang, J.A. Rogers, Soft microfluidic assemblies of sensors, circuits, and radios for the skin, Science 344 (6179) (2014) 70–74, http://dx.doi.org/10.1126/science.1250169.

[7] A.P. Gerratt, N. Sommer, S.P. Lacour, A. Billard, Stretchable capacitive tactile skin on humanoid robot fingers - First experiments and results, in: Proc. of the IEEE Intl. Conf. on Humanoid Robots (Humanoids), 2014, pp. 238–245, http://dx.doi.org/10.1109/HUMANOIDS.2014.7041366.

[8] A. Billard, D. Kragic, Trends and challenges in robot manipulation, Science 364 (6446) (2019) eaat8414, http://dx.doi.org/10.1126/science.aat8414.

[9] F. Liu, F. Sun, B. Fang, X. Li, S. Sun, H. Liu, Hybrid robotic grasping with a soft multimodal gripper and a deep multistage learning scheme, IEEE Trans. Robot. (TRO) 39 (3) (2023) 2379–2399, http://dx.doi.org/10.1109/TRO.2023.3238910.

[10] M. Oller, J. Wu, Z. Wu, J.B. Tenenbaum, A. Rodriguez, See, feel, act: Hierarchical learning for complex manipulation skills with multisensory fusion, Science Robotics 4 (26) (2019) eaav3123, http://dx.doi.org/10.1126/scirobotics.aav3123.

[11] I. Huang, Y. Narang, C. Eppner, B. Sundaralingam, M. Macklin, R. Bajcsy, T. Hermans, D. Fox, DefGraspSim: Physics-based simulation of grasp outcomes for 3D deformable objects, IEEE Robot. Autom. Lett. (RA-L) 7 (3) (2022) 6274–6281, http://dx.doi.org/10.1109/LRA.2022.3158725.

[12] W. Xie, Z. Yu, Z. Zhao, B. Zuo, Y. Wang, HMDO: Markerless multi-view hand manipulation capture with deformable objects, Graph. Models 127 (2023) 101178, http://dx.doi.org/10.1016/j.gmod.2023.101178.

[13] X. Li, Y. Guo, Y. Tu, Y. Ji, Y. Liu, J. Ye, C. Zheng, Textureless deformable object tracking with invisible markers, IEEE Trans. Pattern Anal. Mach. Intell. (TPAMI, Early Access) (2024) 1–12., http://dx.doi.org/10.1109/TPAMI.2024.3463422.

[14] J. Bohg, K. Hausman, B. Sankaran, O. Brock, D. Kragic, S. Schaal, G.S. Sukhatme, Interactive perception: Leveraging action in perception and perception in action, IEEE Trans. Robot. (TRO) 33 (6) (2017) 1273–1291, http://dx.doi.org/10.1109/TRO.2017.2721939.

[15] H. Duan, Y. Yang, D. Li, P. Wang, Human–robot object handover: Recent progress and future direction, Biomim. Intell. Robot. 4 (1) (2024) 100145, http://dx.doi.org/10.1016/j.birob.2024.100145.

[16] M.T. Mason, Toward robotic manipulation, Annu. Rev. Control. Robot. Auton. Syst. 1 (2018) 1–28, http://dx.doi.org/10.1146/annurev-control-060117-104848.

[17] S. Funabashi, T. Isobe, F. Hongyi, A. Hiramoto, A. Schmitz, S. Sugano, T. Ogata, Multi-fingered in-hand manipulation with various object properties using graph convolutional networks and distributed tactile sensors, IEEE Robot. Autom. Lett. (RA-L) 7 (2) (2022) 2102–2109, http://dx.doi.org/10.1109/LRA.2022.3142417.

[18] O. Kroemer, S. Niekum, G. Konidaris, A review of robot learning for manipulation: Challenges, representations, and algorithms, J. Mach. Learn. Res. 22 (30) (2021) 1–82, http://dx.doi.org/10.5555/3546258.3546288.

[19] A. Palleschi, F. Angelini, C. Gabellieri, D.W. Park, L. Pallottino, A. Bicchi, M. Garabini, Grasp it like a pro 2.0: A data-driven approach exploiting basic shape decomposition and human data for grasping unknown objects, IEEE Trans. Robot. (TRO) 39 (5) (2023) 4016–4036, http://dx.doi.org/10.1109/TRO.2023.3286115.

[20] F. Ceola, L. Rosasco, L. Natale, RESPRECT: Speeding-up multi-fingered grasping with residual reinforcement learning, IEEE Robot. Autom. Lett. (RA-L) 9 (4) (2024) 3045–3052, http://dx.doi.org/10.1109/LRA.2024.3363532.

[21] J. Orbik, A. Agostini, D. Lee, Inverse reinforcement learning for dexterous hand manipulation, in: Proc. of the IEEE Intl. Conf. on Development and Learning (ICDL), 2021, pp. 1–7, http://dx.doi.org/10.1109/ICDL49984.2021.9515637.

[22] I. Radosavovic, X. Wang, L. Pinto, J. Malik, State-only imitation learning for dexterous manipulation, in: Proc. of the IEEE/RSJ Intl. Conf. on Intelligent Robots and Systems, IROS, 2021, pp. 7865–7871, http://dx.doi.org/10.1109/IROS51168.2021.9636557.

[23] A. Rajeswaran, V. Kumar, A. Gupta, G. Vezzani, J. Schulman, E. Todorov, S. Levine, Learning complex dexterous manipulation with deep reinforcement learning and demonstrations, in: Proc. of Robotics: Science and Systems, RSS, 14, Pittsburgh, Pennsylvania, 2018, http://dx.doi.org/10.15607/RSS.2018.XIV.049.

[24] L. Shao, T. Migimatsu, Q. Zhang, K. Yang, J. Bohg, Concept2Robot: Learning manipulation concepts from instructions and human demonstrations, Int. J. Robot. Res. (IJRR) 40 (12–14) (2021) 1419–1434, http://dx.doi.org/10.1177/02783649211046285.

[25] Y. Qin, Y.-H. Wu, S. Liu, H. Jiang, R. Yang, Y. Fu, X. Wang, DexMV: Imitation learning for dexterous manipulation from human videos, in: Proc. of the Europ. Conf. on Computer Vision, ECCV, 2022, pp. 570–587, http://dx.doi.org/10.1007/978-3-031-19842-7_33.

[26] S. Bahl, R. Mendonca, L. Chen, U. Jain, D. Pathak, Affordances from human videos as a versatile representation for robotics, in: Proc. of the IEEE Conf. on Computer Vision and Pattern Recognition, CVPR, 2023, pp. 01–13, http://dx.doi.org/10.1109/CVPR52729.2023.01324.

[27] H. Yin, A. Varava, D. Kragic, Modeling, learning, perception, and control methods for deformable object manipulation, Sci. Robot. 6 (54) (2021) eabd8803, http://dx.doi.org/10.1126/scirobotics.abd8803.

[28] Z. Yang, Y. Wang, Y. Jiang, H. Zhang, C. Yang, DeformerNet based 3D deformable objects shape servo control for bimanual robot manipulation, in: Proc. of the IEEE Intl. Conf. on Industrial Technology, ICIT, 2024, pp. 1–7, http://dx.doi.org/10.1109/ICIT58233.2024.10570080.

[29] M. Yu, K. Lv, H. Zhong, S. Song, X. Li, Global model learning for large deformation control of elastic deformable linear objects: An efficient and adaptive approach, IEEE Trans. Robot. (TRO) 39 (1) (2023) 417–436, http://dx.doi.org/10.1109/TRO.2022.3200546.

[30] S. Li, Z. Huang, T. Chen, T. Du, H. Su, J.B. Tenenbaum, C. Gan, DexDeform: Dexterous deformable object manipulation with human demonstrations and differentiable physics, in: Proc. the Intl. Conf. on Learning Representations, ICLR, 2023, https://openreview.net/forum?id=LIV7-_7pYPl.

[31] R. Wu, C. Ning, H. Dong, Learning foresightful dense visual affordance for deformable object manipulation, in: Proc. of the IEEE Intl. Conf. on Computer Vision, ICCV, 2023, pp. 10913–10922, http://dx.doi.org/10.1109/ICCV51070.2023.01005.

[32] Y. Xu, W. Wan, J. Zhang, H. Liu, Z. Shan, H. Shen, R. Wang, H. Geng, Y. Weng, J. Chen, T. Liu, L. Yi, H. Wang, UniDexGrasp: Universal robotic dexterous grasping via learning diverse proposal generation and goal-conditioned policy, in: Proc. of the IEEE Conf. on Computer Vision and Pattern Recognition, CVPR, 2023, pp. 4737–4746, http://dx.doi.org/10.1109/CVPR52729.2023.00459.

[33] T. Zhu, R. Wu, X. Lin, Y. Sun, Toward human-like grasp: Dexterous grasping via semantic representation of object-hand, in: Proc. of the IEEE Intl. Conf. on Computer Vision, ICCV, 2021, pp. 15721–15731, http://dx.doi.org/10.1109/ICCV48922.2021.01545.

[34] T. Zhu, R. Wu, J. Hang, X. Lin, Y. Sun, Toward human-like grasp: Functional grasp by dexterous robotic hand via object-hand semantic representation, IEEE Trans. Pattern Anal. Mach. Intell. (TPAMI) 45 (10) (2023) 12521–12534, http://dx.doi.org/10.1109/TPAMI.2023.3272571.

[35] S. Cui, R. Wang, J. Wei, F. Li, S. Wang, Grasp state assessment of deformable objects using visual-tactile fusion perception, in: Proc. of the IEEE Intl. Conf. on Robotics & Automation, ICRA, 2020, pp. 538–544, http://dx.doi.org/10.1109/ICRA40945.2020.9196787.

[36] R. Gao, Z. Si, Y.-Y. Chang, S. Clarke, J. Bohg, L. Fei-Fei, W. Yuan, J. Wu, ObjectFolder 2.0: A multisensory object dataset for Sim2Real transfer, in: Proc. of the IEEE Conf. on Computer Vision and Pattern Recognition, CVPR, 2022, pp. 10588–10598, http://dx.doi.org/10.1109/CVPR52688.2022.01034.

[37] M.A. Lee, Y. Zhu, K. Srinivasan, P. Shah, S. Savarese, L. Fei-Fei, A. Garg, J. Bohg, Making sense of vision and touch: Self-supervised learning of multimodal representations for contact-rich tasks, in: Proc. of the IEEE Intl. Conf. on Robotics & Automation, ICRA, 2019, pp. 8943–8950, http://dx.doi.org/10.1109/ICRA.2019.8793485.

[38] M.A. Lee, Y. Zhu, P. Zachares, M. Tan, K. Srinivasan, S. Savarese, L. Fei-Fei, A. Garg, J. Bohg, Making sense of vision and touch: Learning multimodal representations for contact-rich tasks, IEEE Trans. Robot. (TRO) 36 (3) (2020) 582–596, http://dx.doi.org/10.1109/TRO.2019.2959445.

[39] H. Liang, L. Cong, N. Hendrich, S. Li, F. Sun, J. Zhang, Multifingered grasping based on multimodal reinforcement learning, IEEE Robot. Autom. Lett. (RA-L) 7 (2) (2022) 1174–1181, http://dx.doi.org/10.1109/LRA.2021.3138545.

[40] F. Ceola, E. Maiettini, L. Rosasco, L. Natale, A grasp pose is all you need: Learning multi-fingered grasping with deep reinforcement learning from vision and touch, in: Proc. of the IEEE/RSJ Intl. Conf. on Intelligent Robots and Systems, IROS, 2023, pp. 2985–2992, http://dx.doi.org/10.1109/IROS55552.2023.10341776.

[41] H. Li, Y. Zhang, J. Zhu, S. Wang, M.A. Lee, H. Xu, E. Adelson, L. Fei-Fei, R. Gao, J. Wu, See, hear, and feel: Smart sensory fusion for robotic manipulation, in: Proc. of the Conf. on Robot Learning (CoRL), 2023, pp. 1368–1378, https://proceedings.mlr.press/v205/li23c.html.

[42] S. Kadalagere Sampath, N. Wang, H. Wu, C. Yang, Review on human-like robot manipulation using dexterous hands, Cogn. Comput. Syst. 5 (1) (2023) 14–29, http://dx.doi.org/10.1049/ccs2.12073.

[43] Y. Liu, Y.-Y. Tsai, B. Huang, J. Guo, Virtual reality based tactile sensing enhancements for bilateral teleoperation system with in-hand manipulation, IEEE Robot. Autom. Lett. (RA-L) 7 (3) (2022) 6998–7005, http://dx.doi.org/10.1109/LRA.2022.3161711.

[44] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: Proc. of the IEEE Conf. on Computer Vision and Pattern Recognition, CVPR, 2016, pp. 770–778, http://dx.doi.org/10.1109/CVPR.2016.90.

[45] Z. Fu, T.Z. Zhao, C. Finn, Mobile ALOHA: Learning bimanual mobile manipulation with low-cost whole-body teleoperation, in: Proc. of the Conf. on Robot Learning (CoRL), 2024, in press, https://mobile-aloha.github.io/.

[46] G. Bertasius, H. Wang, L. Torresani, Is space-time attention all you need for video understanding? in: Proc. of the Intl. Conf. on Machine Learning, 2021, pp. 813–824, https://proceedings.mlr.press/v139/bertasius21a.html.